

Impact of Induced Interpersonal Synchrony on Affiliation Towards Virtual Agent in Human–Agent Interaction

A Thesis (Pre-)Proposal

Submitted to the Department of Informatics of TUM

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1 Introduction

This thesis will explore interpersonal synchrony during human-agent interaction and its possible implications on agent design.

1.1 Background

Only 30% of the communication that happens during human-human interaction is verbal. The other 70% consist of non-verbal behavior and cues that are usually only perceived unconsciously. One of those concepts is interpersonal synchrony, where dyads or groups of persons have been observed to synchronize behavior and movement like body posture or facial expression. This synchrony typically leads to affiliation, meaning that persons who are more in sync report to find each other more likable, even if the synchrony was intentionally induced by one or more of the parties.

1.2 Problem diagnosis

With the rise of technologies like AR and VR, as well as other virtual environments like video games, much of that non-verbal communication gets lost, as virtual agents or avatars become the interface for many interactions. Those agents are typically less responsive and don't react in a human-like manner. If we knew whether these problems can be improved upon by the use of psychological concepts like synchrony, we could create better user interfaces that feel more natural and likable.

1.3 Research question/Thesis statement/Hypothesis

The goal of this thesis is to find out how the positive effects of synchrony of body posture and facial expression can be used to increase affiliation and rapport towards an agent in a human-agent interaction given a typical video game dialogue interaction scenario.

1.4 Methods

First, I will create a virtual humanoid agent that has a basic idle animation. Then I will make use of different sensors to translate a person's body posture and facial expression into animations for a humanoid model. These dynamic animations will be blended with the idle animation of the virtual agent. The blending will be adjustable to test different settings independently. Afterwards, I

plan to conduct a user study with between-subject design with different settings of the given blending, ranging from 0% to 100%. The results will be evaluated using statistical methods like ANOVA.

1.5 Preview

After the introduction, the reader will find an overview of the underlying concepts, the creation of the virtual agent, the experimental procedure, the results, an evaluation of the results together with a comparison to related work and finally a conclusion.

2 Literature review

Many experiments regarding interpersonal synchrony have been conducted so far, each focusing on different facets of synchrony, like Hove & Risen [1] concentrating on rhythmic tapping or Ramseyer & Tschacher [2] analyzing impact of synchrony in a psychotherapeutic setting. Similar experiments have been conducted in the field of human–computer interaction. Experiments by Fujiwara et al. [3] regarding interactions with avatars showed that synchrony with avatars can happen, and that it has similar effects to when it happens between two persons. Another experiment done by Roth et al. [4] injected mimicry—a related concept to synchrony—into a hybrid avatar agent during interaction, successfully inducing known effects of mimicry.

However, to my knowledge, it has not yet been tested whether known effects of synchrony can be observed with a virtual agent when the synchrony is intentionally induced by the agent.

3 Methods

3.1 Implementation

I will implement a virtual humanoid agent with a basic idle animation. The agent's body posture and facial expressions will be blended with the perceived human interactor's body posture and facial expressions. The information of the human interactor will be retrieved by the help of technologies like the HoloLens, the Kinect, basic cameras and potentially other AR/VR hardware. The agent will

have adjustable parameters that decide the amount of synchrony it will display, ranging from 0% to 100%.

3.2 User Study

The agent will be tested in a user study with varying settings for the synchrony percentage with between-subject design. Ideally, I am going to test 5 different settings: 0%, 25%, 50%, 75% and 100%. Each setting would need ten to fifteen testers to be able to formulate a result with impactful meaning, which means I would need around 50 to 75 testers as a minimum requirement. Participants of the user study will be asked to take part in a typical video game interaction scenario with the agent. Afterwards, they will be asked to fill out a form asking about their demographics and a rating of the agent on different attributes for a quantitative measurement. The most important attributes are friendliness and human-likeness.

3.3 Techniques of Analysis

The user study will be evaluated using typical statistical techniques for evaluation like ANOVA.

3.4 Critique

One major problem of the method is the fact that the induced synchrony happens at all times, which is rather uncommon. I might be able to bypass this problem by activating and deactivating the blending based on another parameter, but this might also become problematic, since the transition between no blend and full blend would be very close to mimicry. That could have an impact on the result of the user study, since mimicry can have similar effects on affiliation.

4 Delimitations of the Thesis

There are several items related to this work that I will not address. First of all, interpersonal synchrony is a huge topic with many different layers. Body posture and facial expression are only a few of the forms of synchrony. Furthermore, the work raises ethical questions regarding behavioral modification, which will not be discussed by the thesis, since the discussion itself would warrant a complete thesis and is therefore out of scope.

5 Thesis Schedule

1. Prepare and familiarize + write thesis, 15.04.~10.05.
2. Develop synchrony agent + write thesis, 10.05.~20.06.
3. Set-up user study environment + write thesis, 20.06.~10.07.
4. Conduct user study + write thesis, 10.07.~01.09.
5. Analyze results + write thesis, 01.09.~01.10.
6. Finalize thesis, 01.10.~15.10.

6 Bibliography

- [1] Hove and Risen, It's all in the timing: Interpersonal synchrony increases affiliation, *Social cognition*, 27/6, 2009.
- [2] Ramseyer and Tschacher, Nonverbal synchrony of head- and body-movement in psychotherapy: different signals have different associations with outcome, *Frontiers in psychology*, 5, 2014.
- [3] Fujiwara et al., Synchrony facilitates altruistic decision making for non-human avatars, *Computers in Human Behavior*, 128, 2022.
- [4] Roth et al., Injecting nonverbal mimicry with hybrid avatar-agent technologies: A naive approach, *Proceedings of the Symposium on Spatial User Interaction*, 2018.
- [5] Rennung and Göritz, Prosocial consequences of interpersonal synchrony, *Zeitschrift für Psychologie*, 2016.
- Meta analysis of 60 different synchrony experiments.
- [6] Gratch et al., Can virtual humans be more engaging than real ones?, *International Conference on Human-Computer Interaction*, p. 286–297, 2007.
- [7] Gupta, Aman, et al., Blink as you sync: uncovering eye and nod synchrony in conversation using wearable sensing, *Proceedings of the 23rd International Symposium on Wearable Computers*. 2019.
- [8] Miller et al., Social interaction in augmented reality., *PloS one*, 2019.
- [9] Fraser et al., Spoken conversational AI in video games: Emotional dialogue management increases user engagement, *Proceedings of the 18th International Conference on Intelligent Virtual Agents*, 2018.

7 Appendices

Beyond the written thesis, some other items will be produced:

- _ Synchrony agent with adjustable synchrony parameter
- _ Experimental data